

Alicia Sagae, Ph.D.

RESEARCH SCIENTIST IN NLP/RESPONSIBLE AI

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Alicia Sagae is a scientist in Responsible AI, computational linguistics, and natural language processing. Alicia brings precision, enthusiasm, work ethic, and strong collaboration skills to every project. She has over 15 years of experience building industrial and academic NLP systems, both in-office and remotely. Alicia has programmed and run projects in every phase of development, from data acquisition to modeling, evaluation, reporting, and communicating impact.

Experience

Amazon AWS

Davis, CA (Remote)

RESEARCH SCIENTIST IN ASR/NLP (RESPONSIBLE AI)

2021 - PRESENT

- Design, code, and execute Responsible AI assessments of existing AI services. Deliver technical reports and review recommendations with leadership/service developers
- Collaborate with leadership to prioritize projects in Responsible AI. Deliver code, diagrams, task lists, and work plans
- Develop, code, and present hands-on builder workshops on RAI topics at flagship events (AWS re:Invent)
- Contribute to public-facing RAI processes and documentation for AI services

Alelo, Inc.

Los Angeles, CA (Remote)

PRINCIPAL ARTIFICIAL INTELLIGENCE ENGINEER

2018-2021

- Technical leadership in Artificial Intelligence, Machine Learning, and Data Management for software products supporting real-time spoken conversation between students and AI characters
- Migrate legacy dialog manager to a machine-learned solution based on Microsoft Azure Luis
- Implement custom quality assurance tests and performance tracking for conversational AI
- Write custom experiments to validate learning outcomes

Consulting Scientist in Natural Language Processing (NLP)

Davis, CA

INDEPENDENT CONTRACTOR

2014 - 2021

- Design, code, and evaluate NLP solutions for companies in the educational technology sector
- Collaborate with managers and CEO/CTO to crystallize a vision for AI software. Deliver software engineering artifacts based on this vision (code, diagrams, task lists, work plans)
- Strengthen communication channels in remote interdisciplinary teams
- Measure success and report outcomes across teams, in terms they care about

E Touch Systems Corp.

Fremont, CA (Remote)

SENIOR RESEARCH SCIENTIST

2014

- Consultant on assignment to Google. Build and evaluate a system to classify customer tickets automatically and reduce manual effort to resolve them

Sample Projects

AI Service Card for Amazon Transcribe

- Science lead for bias assessment of an existing AI service for Automatic Speech Recognition (ASR)
- With teammates, design speech datasets with annotations that expose dimensions of bias in speech systems
- Select bias metrics for ASR on this data; code and execute metrics, report findings
- With leadership, design a public-facing AI Service Card that addresses RAI issues including bias: <https://tinyurl.com/asrcard>
- Work directly with customer ethics teams to certify AWS RAI practices, based on this experimental evidence and on the Service Card
- **Impact:** Industry-leading transparency on bias in AI systems. Increased customer trust leading directly to new contracts

Alelo Enskill

- AI lead for a conversational simulation platform for learners of English practicing their speaking skills
- Migrate existing dialogue manager from rule-based to machine-learned algorithm. Direct an engineering team responsible for updating existing code. Find 3rd party components and write custom code in Python and JavaScript
- Train team members that use the new dialogue manager: content authors, engineers, and project managers
- **Impact:** Improve intent analysis by 20%-50%, depending on simulation

Learner Modeling

- Redesign learner logs for better student skill tracking in a system for language instruction. Pull data and validate logging infrastructure using MongoDB, Python, and Perl
- Create and tune mathematical formulas for scoring student speech in real time
- Validate software effectiveness using the new logs and formulas. Sample student groups, track skill over time, and report learning outcomes
- **Impact:** New capabilities to track learner behavior over time and report fluency scores in real time. Increased visibility, credibility and buy-in with teachers and students

Product Translation

- Create an end-to-end work plan to translate an existing conversational AI simulation from English into South American Spanish
- Lead the data development: select data, create guidelines, meet with translators, and review quality
- Retrain conversational AI models using Spanish data, re-test, tune, report performance, and suggest further improvements
- **Impact:** Product is available in new markets

CultureCom

- Serve as Principal Investigator. Write and manage technical goals, milestones, government deliverables. 3 years/\$2M budget
- Lead the team to deliver a system for resolving logical constraints in dialogue. Model social constraints in simulated dialogue between a player and an AI character as logical forms
- **Impact:** Add new capability to the conversational simulation. Drive company profit by acquiring the project and managing it on-budget. Expand company presence in relevant research communities through scientific publication

Education

Carnegie Mellon University

PH.D. LANGUAGE AND INFORMATION TECHNOLOGIES

Pittsburgh, PA

2010

- **Tokyo University "Todai"**, Tokyo, Japan — *Visiting Researcher* 2006-2008

- **University of Karlsruhe**, Karlsruhe, Germany — *Visiting Researcher* 2002-2003

University of North Carolina

Chapel Hill, NC

B.SCI. MATHEMATICAL SCIENCES, B.A. SLAVIC LINGUISTICS (DUAL DEGREE)

2000

Selected Publications

ARTICLES AND CHAPTERS

Image Retrieval with Textual Label Similarity Features

Alicia Sagae, Scott E. Fahlman

Intelligent Systems in Accounting, Finance, and Management 22.1 (Jan. 2015) pp. 101–113. 2015

Conversational Agents in Language and Culture Training

Alicia Sagae, W. Lewis Johnson, Andre Valente

Conversational Agents and Natural Language Interaction: Techniques and Effective Practices, 2011, Madrid

CONFERENCE PROCEEDINGS

A Use-Case Specific Dataset for Measuring Dimensions of Responsible Performance in LLM-generated Text

Alicia Sagae, Chia-Jung Lee, Sandeep Avula, Brandon Dang, Vanessa Murdock

Proceedings of the 34th ACM International Conference on Information and Knowledge Management, 2025, Seoul, Republic of Korea

Knowledge Resource Development for Identifying Matching Image Descriptions

Alicia Sagae, Scott E. Fahlman

5th International Conference on Knowledge Engineering and Ontology Development (KEOD 2013) acceptance rate 13%, 2013

Efficient Cultural Models of Verbal Behavior for Communicative Agents

Alicia Sagae, Jerry R. Hobbs, Suzanne Wertheim, Michael Agar, Emily Ho, W. Lewis Johnson

12th International Conference on Intelligent Virtual Agents (IVA 2012), 2012

Adaptive Agent-Based Learning of Intercultural Communication Skills

W. Lewis Johnson, Alicia Sagae

User Modeling and Personalization Conference, Demonstration Sessions (UMAP 2010), 2010, Tokyo, Japan

Validation of a Dialog System for Language Learners

Alicia Sagae, W. Lewis Johnson, Stephen Bodnar

11th annual SIGdial Meeting on Discourse and Dialogue, 2010, Tokyo, Japan

A Comparison of Knowledge Resource Designs: Supporting Term-Level Text Annotation

Alicia Tribble, Jin-Dong Kim, Tomoko Ohta, Jun'ichi Tsujii

LREC-2008 Workshop: Building and evaluating resources for biotext mining, 2008, Marrakesh, Morocco

CMU-AT: Semantic Distance and Background Knowledge for Identifying Semantic Relations

Alicia Tribble, Scott E. Fahlman

SemEval-2007 Workshop of the meeting of the Association for Computational Linguistics (ACL 2007), 2007, Prague, Czech Republic

SconeEdit: A Text-Guided Domain Knowledge Editor

Alicia Tribble, Benjamin Lambert, Scott E. Fahlman

Demonstration Sessions of the Human Language Technologies Conference (HLT/NAACL-2006), 2006, New York, NY

Overlapping Phrase-level Translation Rules in an SMT Engine

Alicia Tribble, Stephan Vogel, Alex Waibel

Proceedings of NLPKE-2003 Natural Language Processing and Knowledge Engineering, 2003, Beijing, China

Speaker, Accent, and Language Identification Using Multilingual Phone Strings

Tanja Schultz, Qin Jin, Kornel Laskowski, Alicia Tribble, Alex Waibel

Proceedings of HLT-2002 Human Language Technology Conference, 2002, San Diego, California

Improving Statistical Machine Translation for a Speech-to-Speech Translation Task

Stephan Vogel, Alicia Tribble

Proceedings of ICSLP-2002 Workshop on Speech-to-Speech Translation, 2002, Boulder, CO